Shilpa Thakur

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*Seeking co-op and internship opportunities from Jan 2026

Education

Worcester Polytechnic Institute (WPI), MA, USA
Ph.D., Robotics Eng. focused on Human-Machine Interaction, Interfaces and Intelligence, **Jan 2022 – Present**
SRM Institute of Science and Technology Kattankulathur, Tamilnadu, India
M Tech. Robotics Eng., Gold Medalist, (9.8/10), **Jul 2018 – Jul 2020**
B. Tech. Mechatronics, (8.7/10), **Jul 2014 – Jul 2018**

Certifications

Certified SolidWorks Professional (CSWP) Mechanical Designer, Dassault Systems, **Dec 2019**
Robotics Engineering, NPTEL (IIT Kharagpur), **Sept 2018**

Skill Set.

Computer Skills : Python, C++, MATLAB & Simulink, Git Workflow, Git Version Control, Open CV, Blender API, SolidWorks, COMSOL, KiCAD EDA, Linux (or Unix) Terminal / Shell, Robot Operating System (ROS), MS Office (Word, PowerPoint, Excel)
Lab / Workshop Tools: Oscilloscopes (DSO, MSO, CRO), Arbitrary Waveform Generator (AWG), Function Generator (FG), Digital Multi Meter (DMM), Programmable Power Supply, Programmable Electronic Loads, FDM and SLA 3D Printers, Drills, Milling Machines, Angle Grinders, Lathes, SMD and Through Hole Soldering Stations.

Experience

- Research Assistant, Soft Robotics Lab, WPI**, (Aug 2022- Present)**
- Teaching Assistant, Robotics Engineering Dept., WPI (**Aug 2024- May 2025)**
- Graduate Mentor for Women in Research and Mentorship Program, WPI **(Jan 2023- Sep 2023)**
- Industry Mentor under La Foundation's Connect Next Program, Dassault Systemes, India **(Jan 2022- Aug 2022)**
- Mechatronics Engineer and Co-founder, Incoen Systems, Tamilnadu, India **(Oct 2020-Nov 2021)**

Relevant Projects

- **Human Hand Motion Capture Glove with Embedded Inductive Sensing Soft Haptic Muscle and ML Gesture Recognition (Aug 2024 – Present)**
  - 3D CAD modeling and electro-magnetic FEA simulations to optimize inductive sensor design for capturing the flexion of fingers (SolidWorks, COMSOL)
  - Developing electronics hardware and embedded firmware for inductive sensing-based hand motion capture and haptic feedback rendering with calibration routines, data filtering, mapping and ROS interface nodes for robotic teleoperation or AR environment interaction.
  - Developing a ML engine for hand gesture and sign language recognition used for triggering programmed or learnt routines for teleoperation.
- **Visual-Inertial Odometry Framework using Classical & Deep Learning Approaches (April 2025 – May 2025)**
  - Implemented a stereo VIO system using the Multi-State Constraint Kalman Filter (MSCKF) for accurate pose estimation in GPS-denied environments.
  - Developed deep learning models for visual, inertial, and fused VIO, trained on synthetic drone trajectories to enhance robustness in dynamic conditions.
- **Blender-Based Visual Perception System for Self-Driving Cars (March 2025 – April 2025)**
  - Designed a Blender-based visual perception framework simulating autonomous driving scene understanding with real-time overlays for objects, lanes, depth, and motion.
  - Integrated state-of-the-art computer vision models (YOLO, Owl-ViT, LaneNet, DepthPro, OpenPose, RAFT) for open-vocabulary detection, lane segmentation, depth estimation, and dynamic object tracking.
- **Teleoperation with Integrated Learning from Demonstration Feature (Apr 2023 – Sep 2024)**

Project team of 5. Amazon Robotics funded collaboration between WPI, Harvard and University of Massachusetts Lowell.
  - Developed a three fingered tendon driven gripper for Kinova Gen 3 articulated robot with multiple 3D hall-effect based contact force sensors.
  - Developed the MCU firmware and the I2C communication hub for interfacing, calibrating, sampling and filtering all contact force sensors.
  - Developed the ROS node for the custom gripper and integrated it with the team's robotic teleoperation framework.
  - Integrated the wearable tether-less human machine interface backpack with human arm motion capture and haptics (developed previously) for data collection.
- **Wearable human machine interface (HMI) backpack with full arm motion capture and haptics for robot teleoperation (Aug 2022-March 23)**

Project team of 5. SEMI FlexTech, US Army Research Lab (ARL) and National Science Foundation (NSF) funded project, WPI.
  - Designed and developed a wearable HMI backpack integrating electronics, pneumatics, power systems, and compressor for full-arm motion capture and haptic feedback (SolidWorks, KiCAD).
  - Developed I2C IMU network with forward kinematics for spatial tracking, Hall-effect force sensor (≤1 Kg) on Kinova Gen 3 gripper, and pneumatic soft actuators for tactile feedback.
  - Implemented MCU firmware and ROS nodes for real-time sensing, haptic control, teleoperation interface, and data logging/visualization. (C, Python, Matplotlib)
- **Two-stage electro-pneumatic delta style industrial robot and custom robot controller (July – Oct 2018)**
  - Designed and simulated a multi-stage electro-pneumatic delta manipulator with cascaded electric and pneumatic stages (SolidWorks CAD & motion study).
  - Built a custom robot controller integrating MCU, motor drivers, solenoid drivers, sensors, and full power management circuitry.
  - Programmed firmware for kinematics, closed-loop control, system health monitoring and pneumatic pressure regulation.

Publications

- "Soft RING : Realtime Inductive Glove for Hand Motion Capture and Haptic Feedback", (Submitted at IEEE Transactions on Robotics (T-RO) Journal), Sept 2025
- "A Tetherless Soft Robotic Wearable Haptic Human Machine Interface for Robot Teleoperation", IEEE RSJ International Conference on Intelligent Robots and Systems (IROS), Oct 2024
- "Braided Composite System with Haptic Feedback for Teleoperation", Polymeric Processing Society (PPS) 2024.

- "Substrate Surface Carburization-Based Electro-Mechanical Plotter Using Plasma Evolved During Electrostatic Breakdown of Ambient Air", Springer - Journal of Thermal Analysis and Calorimetry, Feb 2021
- "A Novel Three Axis Semi Elliptic Manipulator Configuration for Simultaneous Multi Point Operations", IOP Science Conference Series – Materials Science and Engineering, April 2020

Patents

Servo ERM micro actuator (201841017350), Inkless Printing using Surface Plasma Carburization (201841036362), Helical Electrode Drum for High-Speed Inkless Printing (201941025781), Closed Loop Plasma Control System for Inkless Plasma Printing (201941028522)

Achievements

- **Jul 2020 : Department 1st Position, Gold Medal,** for academic excellence in Masters of Technology (M. Tech.) in Robotics Engineering.
- **Jul 2019 : National 1st Position, Chairman's Award for Best Innovation & 12,500 USD in Cash,** for "Inkless Plasma Printing Technology" submitted and demonstrated at the Indian Innovation Challenge Design Contest (IICDC) by Texas Instruments, Department of Science and Technology (DST) Govt. of India and the Indian Institute of Management Bangalore (IIMB).
- **Feb 2018 : Best Research Project** of the Mechatronics Department for "Inkless Plasma Printing Technology".
- **Oct 2018 : Best Innovation Award** for developing and demonstrating the "Two-stage Electro-Pneumatic Delta Style Robot and Custom Robot Controller" at International AI and Robotics Expo 2018 by Robotics and AI Foundation of India (RAAIF).